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BIO-INSPIRED ENGINEERING 4: COURSEWORK ASSIGNMENT

Mimicking termite's mounds principles for cooling in buildings

Abstract

In this paper we discuss a new possible way of cooling buildings. First we look at the actual cooling system problems as energy consumption and pollution. Next we study the termites mound searching for inspiration, explaining the mechanism they use to maintain almost constant the temperature. Later we analyse the Eastgate Centre, which uses a cooling system bases on termites. Afterward we explain the crucial feature termite-inspired building design, as gaps in between walls and different materials and how it works. Finally, we discuss the potential benefits and problems that can appear, ending with the future design steps.

Problem-Building cooling

Most people currently spend 90% of their daily lives indoors [1], thus buildings account for over a third of global energy consumption and were responsible for an estimated 28% of global energy-related CO₂ emissions in 2018, [1-4] A high percentage of all that energy consumed in the buildings is used by Heating, Ventilation and Air Conditioning systems (HVAC), from which air cooling account for 10-25 % of total energy consumption [1-2,5-7] and is the fastest growing end-use in buildings, mainly in consequence of the higher living standards from the non- OECD countries (mainly China and India) [2-5],the tendency is shown in figure 1.

All this electric energy comes mostly from burning fossil fuels, causing global warming and ozone layer depletion, ice melting and the emissions of pollutant gases, figure 1 left. There are also other concerns with cooling systems as the cost of the electricity derived from the energy use and the poor quality of the air which causes respiratory problem and necessity of using ventilation systems. A reduction in the energy used for cooling and a more 'natural' system will help to fight against all of these problems.

The traditional conception of building-as-machine [8] is based on active cooling: air conditioning and chillers [6], consuming a huge amount of energy. To switch from the traditional concept in favour of a living homeostatic building: which uses less energy for cooling (or none) thanks to passive cooling systems, making the building as comfortable as with air conditioning but without it; a building which regulates the temperature and ventilates itself naturally. This philosophy has been implanted for some big buildings, but it has not been split in design principles and apply to a more normal family's house.

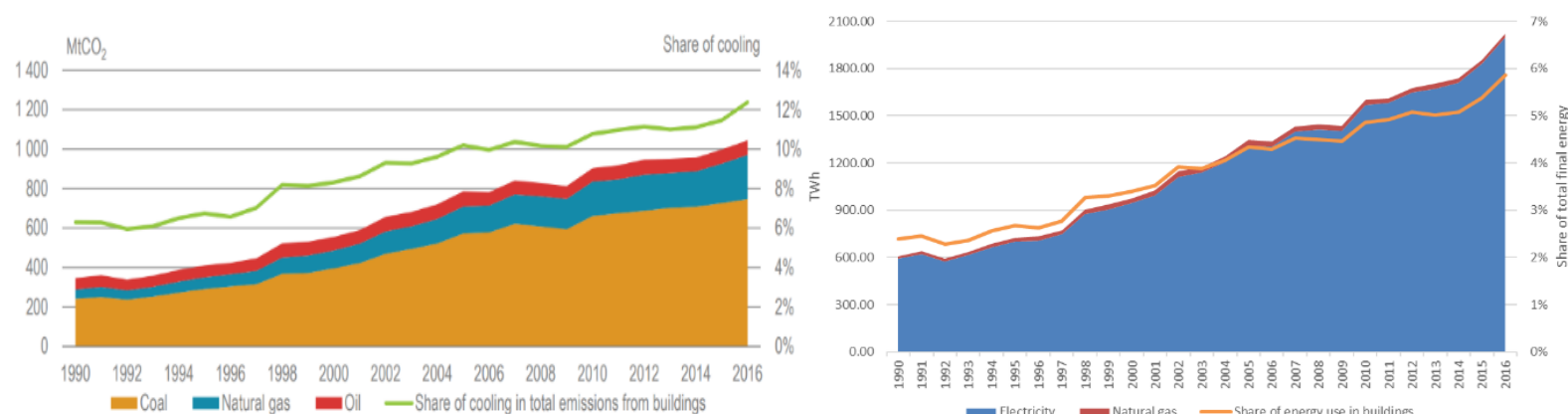


Figure 1: Left: Total emission of CO₂ for cooling purposes and the share of total building emissions from 1990 to 2016 Right: Total energy consumed for cooling purposes and the share of total final energy from 1990 to 2016 [3]

Inspiration- Termite mounds

The termites are social insects that build massive, complex structures which include a subterranean nest and a mound which is relatively uninhabited [9]. The impressive thing of these structures is that they regulate themselves the temperature, the humidity and the CO₂ concentration. We can use the same working principles to inspire passive cooling and ventilation systems. There are two main ventilation drivers in termite's mounds:

- heat induced ventilation The mound geometry consists of flutes and buttress-like structures with a low thermal mass which distributes radially from a large central chimney with a higher thermal mass, figure 2 top left. During the daytime (night-time), the periphery is hotter (colder), while the centre stays cool (warm), creating a convection cell with upwards (downwards) flow in the periphery and downwards (upwards) flow in the centre, figure 2 bottom left- 'midday' ('night'). The mound walls are made of clay with small pores, helping the gases exchange through diffusion [9-11].
- Induced flow theory for winds faster than 3m/s: The upper part of the mound, figure 2 right, exposed to a higher velocity than the openings in lower levels causes the air to rise thanks to Venturi effect, consequently fresh air is drawn into the mound through the lower vents and ventilating the mound [8,11]

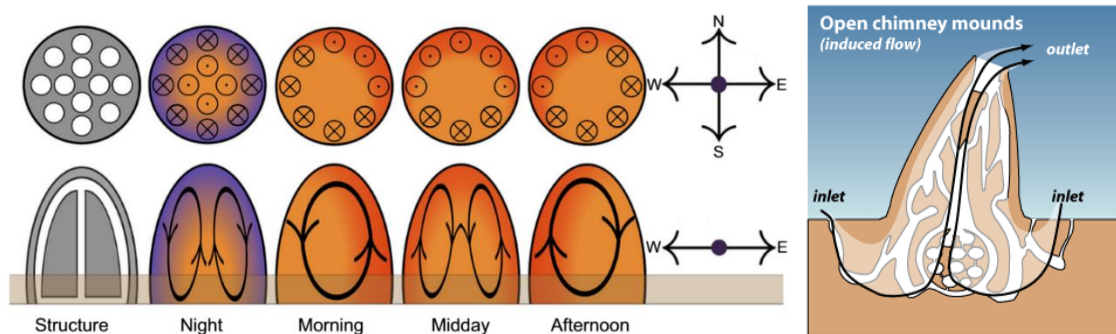


Figure 2: Left: heat induced ventilation with the outer flutes and the central chimney, the flow (black lines) change depending on the side of the mound exposed to sunlight. Right: Flow for induced flow-mounds. [8]

Solution-From Eastgate centre to a family house

The concept of induced flow has been already used in building like the Eastgate Centre, in Harare, Zimbabwe built in 1996, London's Portcullis House, built in 2001 and Council House Two (CH2) in Melbourne, built in 2006, all of them have been designed by the architect Mick Pearce, inspired by termite's mound and using passive cooling [12-13].

In the Eastgate centre, figure 3, the passive cooling works by storing heat in the day and venting it at night as temperatures drop. At the beginning of the day, the building is cool from the night before. During day machines and people generate heat, and the sun shines. Heat is absorbed by the fabric of the building, which has a high heat capacity so that the temperature inside increases but not greatly. In the evening the temperatures outside drop. The warm internal air is vented through chimneys, assisted by low volume fans but also rising naturally because of the stack effect. At night, cold air flowing through cavities in the floor slabs, helped by high volume fans until the building's fabric has reached the ideal temperature to start the next day.

Eastgate uses 35% less total energy than the average consumption of other conventional buildings with full HVAC in Harare. The saving on the capital cost compared with full HVAC was 10% of total building cost [13]. But it still uses fans and it is mainly designed for big office buildings and a modest hot climate, as one can notice from the location of the buildings. Our aim is to subtract the main features of the mound building and apply them to a family's house.

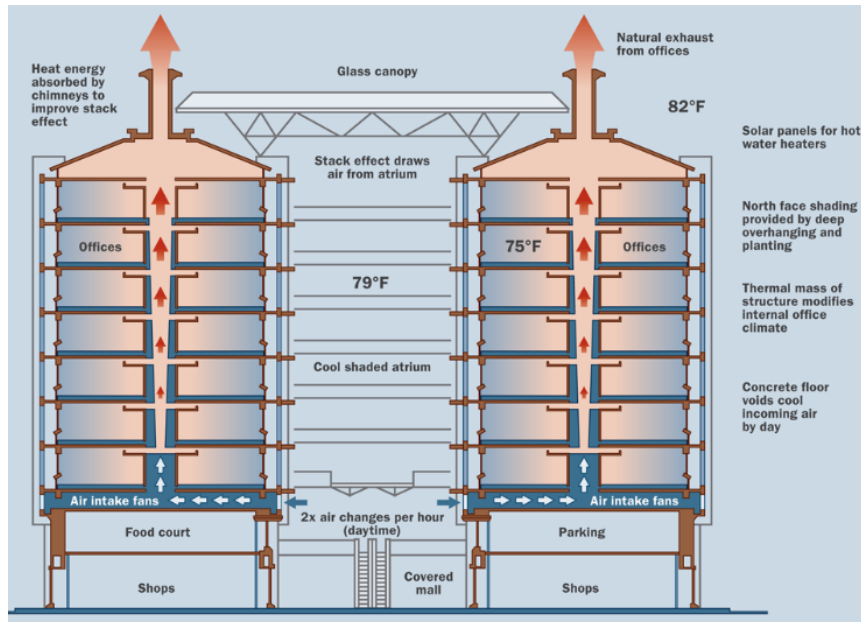


Figure 3: Eastgate Centre's working principles: The air is drawn from the bottom and raised as it heats up (stack effect) [13].

With the termite's principles, we can design a small building with this chimney-flutes design which regulates itself, shown in figure 4.

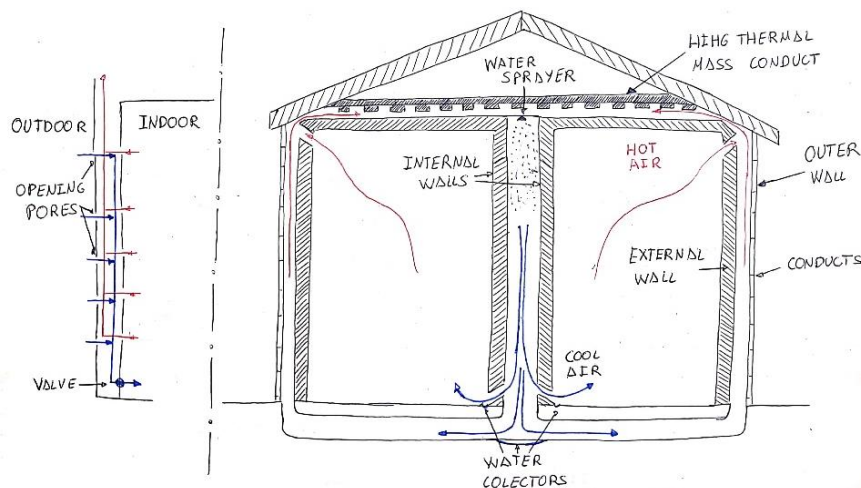


Figure 4: Left: Porous wall to allow gas exchange. Right: Mechanism of convective flow circuit in a small house. Red and blue arrows represent hot and cold air respectively.

The house has a number of crucial design features:

1. The outer wall has a gap, between the external wall and outer wall in figure 4 right, where the air heats as it passes through.
2. The internal and external walls and the top conduits have a high thermal mass material and are made from materials such as concrete. The conduits have some square protuberances, shown in figure 4 right-high thermal mass conduit, to increase the surface and the thermal mass.
3. The outer walls have a low thermal mass and some small conduits to allow the exchange and renewing of air.
4. A net of conduits in between internal wall (chimneys) connected to the outer flutes by the floor and ceiling conduits. Those chimneys can be equipped with a water pulveriser to improve the cooling.
5. In building with an all-day shadow side and for moderate climates it can implement a system with porous walls which allow a natural exchange of gases, figure 4 left.

At the start of the day, the outer walls heat up very quickly, thus heating the air in the conduits, which thanks to the stack effect flow from the floor to the ceiling, so the hotter air of the room and the air in the chimneys are also drawn, as shown in figure 4 right. During this process, the hot air is cool down thanks to the high thermal mass of the wall which are cool from the night before, and the cooling effect of the water's evaporation. The cool air is then introduced in the room's floor. This process occurs during the day and it reverts in the night if the outdoor temperature is cooler than inside or stops if it is hotter (water sprayer required). The water from the pulverization is captured in the bottom and reutilized again.

Evaluation

This kind of construction can improve the energy saving of the building, making the air conditioning unnecessary, thus saving money and producing less pollution for the environment, while maintaining the building cool. It also improves the overall insulation of the building making it useful even in winter when the flow can be stopped to avoid the cooling of the building. The system can be used in any type of building, and have great opportunities for changes in the design as it is based on 5 principal design features. In particular, the outer wall can be used as an aesthetic complement, improving the house appearance as well. Regulating the water pulveriser also allows us to control the humidity of the building.

The gap in between wall is a common practice as it is used for isolating purposes, but the addition of the floor and ceiling conduits and the water pulveriser could cause an increase of the material used and thus of the initial cost of the building. One of the main queries that arise is if that increase in the initial cost of the building will be worth it and how much time it will be payback; this has to be done through a deeper a very careful analysis.

Another concern about the systems if it is will really make the air move through the conduits. For that, we make an initial approximation with stack effect equations (1) to know the velocity of the flow in the closed loop:

$$V = \sqrt{\frac{2 \cdot g \cdot \Delta\rho \cdot h}{\frac{\lambda \cdot l}{d_h} + \sum \varepsilon \cdot \rho_r}} \quad (1)$$

Assuming a model like figure 4 with 100 m² (10x10m), an air gap of 0.1m, a height of 2.5m [14], a friction coefficient of 0.05, a loss coefficient of 1, a floor-air temperature of 20°C and a roof-air temperature of 30°C:

$$V = \sqrt{\frac{2 \cdot 9.81 \cdot 0.04 \cdot 2.5}{\frac{0.05 \cdot 16}{0.1} + 1 \cdot 1.205}} = 0.46m/s \quad (2)$$

This tells us the system can work in principle but a deeper more exhaustive thermo-fluids analysis should be done.

We also need to know how much cooling the system provides and if it actually works. This could be done via thermo-fluids analysis. If a real scale prototype is constructed, a necessary comparison, between the 'homeostatic' house and a normal one, in the actual amount of energy required have to be done. Another question will be the use of water for cooling, so it can create humidity inside the house, thus a proper insulation a collector system must be implemented.

This is just a conceptual design which could potentially solve the problems of cooling family's houses and improve the way we design buildings. The mains questions to work on in a future are: if it will be cheap enough to worth it and the actual amount of flow and cooling power it provides, which must be done in a further design stage.

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